

Notes: Nevo (JEMS, 2000)

A Practitioner's Guide to Estimation of Random-Coefficients Logit Models of Demand

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1 Big picture

- **Question.** How should researchers estimate demand for differentiated products when there are many products, consumers have heterogeneous tastes, and prices are endogenous?
- **Purpose of the paper.** The paper is a practical guide to the Berry (1994) and Berry, Levinsohn, and Pakes (1995) random-coefficients logit demand estimator. It is written to lower the barrier to using these models rather than to introduce a new estimator.
- **Core problem.** A fully flexible demand system for many differentiated products has too many own- and cross-price parameters. A simple logit model solves the dimensionality problem but imposes unrealistic substitution patterns: consumers substitute toward products in proportion to market shares, not toward products with similar characteristics.
- **Main modeling idea.** Project products into a characteristics space and allow consumer tastes for characteristics, including price sensitivity, to differ across consumers. The resulting model keeps the dimensionality manageable while generating richer substitution patterns.
- **Main econometric idea.** The unobserved product characteristic, ξ_{jt} , is the structural error. Since firms observe ξ_{jt} when setting prices, prices are generally endogenous. The BLP/Nevo procedure inverts market shares to recover mean utilities, then uses GMM and instruments to deal with price endogeneity.
- **Data advantage.** The model can be estimated using market-level data: market shares, prices, product characteristics, and optional information about the distribution of consumer demographics. Individual purchase data are useful but not required.
- **Key computational contribution.** For a candidate set of nonlinear taste-distribution parameters, find the mean utility vector δ_t that exactly matches observed shares in each market. Then form the implied residual ξ_{jt} and minimize IV/GMM moments.
- **Practical payoff.** The method produces more plausible own-price and cross-price elasticities. Products that are close in characteristics space become close substitutes, rather than substitution being mechanically pinned down by market shares.
- **Main warning.** The technical machinery does not remove the fundamental identification problem. Credible instruments and credible assumptions about unobserved product quality remain central.
- **Economic takeaway.** Random-coefficients logit is useful because it combines three features: many products, consumer heterogeneity, and price endogeneity. Its value comes from translating market shares into economically meaningful substitution patterns.

2 Data and measurement (key objects)

2.1 Market-level data

The estimator is designed for differentiated-product settings with multiple markets. A market can be a year, a city-quarter, a demographic group, or any environment in which consumers face a product choice set.

- Markets are indexed by $t = 1, \dots, T$.
- Products are indexed by $j = 1, \dots, J$, though the set of available products can vary by market.
- Consumers in market t are indexed by $i = 1, \dots, I_t$.
- The researcher observes market shares, prices, and observed product characteristics for the inside goods.
- The researcher usually does not observe individual purchase decisions.

Examples of market definitions.

- BLP define a market as the national automobile market in a year.
- Nevo's cereal work defines a market as a city-quarter.
- Other applications can define markets by income groups, regions, or time periods.

2.2 Market shares and the outside good

A central data object is the market share of product j in market t , denoted S_{jt} . These shares must be measured relative to a total market size.

- The inside-good shares are based on quantities, such as automobile sales or cereal servings.
- The outside good is the option of not buying any of the observed products.
- The outside share is

$$S_{0t} = 1 - \sum_{j=1}^{J_t} S_{jt}.$$

- The total market size is rarely observed directly and must be chosen using institutional knowledge.

Why this matters.

- Without an outside good, a common increase in all inside-good prices would not reduce total inside-good demand.
- The outside share helps identify the level of price sensitivity and the substitution between inside goods and not purchasing.
- The researcher should choose a market size large enough to leave a nonzero outside share and should check sensitivity to this choice.

2.3 Product characteristics

Observed product characteristics are denoted x_{jt} . They can include physical attributes, quality measures, and segment indicators.

- In automobile applications, examples include horsepower, size, and air conditioning.
- In cereal applications, examples include sugar content, sodium, fiber, and a dummy for whether the product becomes mushy in milk.

- Characteristics can be collected from manufacturer descriptions, trade publications, or other product-level data sources.

Product characteristics have two distinct roles:

1. **Mean utility role.** They explain average consumer valuation of each product.
2. **Substitution role.** They determine which products are close substitutes when consumer tastes for characteristics are heterogeneous.

Important distinction. If there are enough markets, product fixed effects can absorb mean product quality. But product characteristics are still needed to drive substitution patterns. Fixed effects alone do not tell the model that two cereals, cars, or computers are close substitutes.

2.4 Unobserved product characteristics

The researcher observes only some characteristics. The unobserved characteristic is denoted ξ_{jt} .

- Examples include brand equity, unobserved promotions, style, quality, advertising effects, and systematic demand shocks.
- Consumers and firms are assumed to observe these characteristics.
- The econometrician does not observe them, so they enter the demand residual.

The price endogeneity problem comes from this object. Firms set prices after observing ξ_{jt} , so price is generally correlated with the structural demand error.

2.5 Consumer demographics and unobserved heterogeneity

Consumer heterogeneity is modeled using two types of individual characteristics:

- **Observed demographics:** D_i , such as income, age, education, family size, or whether the consumer is a child.
- **Unobserved heterogeneity:** ν_i , such as idiosyncratic taste shifters not measured in external data.

Even when the researcher does not observe the demographics of the actual buyers, information about the distribution of demographics can be brought in from sources such as the Current Population Survey.

Interpretation.

- Demographics reduce reliance on arbitrary parametric distributions.
- If demographic distributions differ across markets, the model can use these differences to identify how tastes vary with demographics.
- Without demographic data, heterogeneity must come mostly from distributional assumptions about ν_i .

3 Model and demand system (all equations + interpretation)

3.1 Indirect utility

Consumer i 's utility from inside good j in market t is

$$u_{ijt} = \alpha_i(y_i - p_{jt}) + x_{jt}\beta_i + \xi_{jt} + \varepsilon_{ijt}, \quad j = 1, \dots, J_t. \quad (1)$$

Objects.

- y_i : consumer income.
- p_{jt} : price of product j in market t .
- x_{jt} : observed product characteristics.
- ξ_{jt} : unobserved product characteristic.
- ε_{ijt} : idiosyncratic taste shock.
- α_i : marginal utility of income, or price sensitivity.
- β_i : taste coefficients for observed characteristics.

Interpretation.

- Consumers differ because α_i and β_i are consumer-specific.
- Products differ because they have different observed characteristics, prices, and unobserved qualities.
- Prices are potentially endogenous because p_{jt} can be correlated with ξ_{jt} .

3.2 Taste heterogeneity

Taste coefficients are modeled as functions of demographics and unobserved random tastes:

$$\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma \nu_i, \quad \nu_i \sim P_\nu^*, \quad D_i \sim P_D^*. \quad (2)$$

Objects.

- Π measures how tastes vary with observed demographics.
- Σ determines the variance and covariance of unobserved random coefficients.
- P_D^* can be empirical, for example from Census or CPS data.
- P_ν^* is often assumed to be multivariate normal.

Plain English. The model lets low-income and high-income consumers have different price sensitivities, children and adults have different taste for sugar, and otherwise similar consumers have idiosyncratic taste differences.

3.3 Mean utility and individual deviation

Substituting equation (2) into equation (1) gives the standard BLP decomposition:

$$u_{ijt} = \alpha_i y_i + \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt}, \quad (3)$$

where

$$\delta_{jt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}, \quad \mu_{ijt} = [-p_{jt}, x_{jt}] (\Pi D_i + \Sigma \nu_i). \quad (4)$$

Interpretation.

- δ_{jt} is the mean utility of product j in market t . It is common to all consumers.
- μ_{ijt} is the individual-specific deviation from mean utility.
- ξ_{jt} enters only through δ_{jt} , which is what makes the inversion strategy possible.
- The income term $\alpha_i y_i$ is common across all inside products and the outside good, so it drops out of choice probabilities after normalization.

3.4 Outside option

The outside good is normalized to have zero utility:

$$u_{i0t} = \varepsilon_{i0t}. \quad (5)$$

This normalization is standard because the mean utility of the outside option and individual-specific constants are not separately identified from the inside-good utilities.

Why the outside good is essential.

- It allows total inside-good demand to fall when all inside-good prices rise.
- It anchors the level of utility and price sensitivity.
- It makes market size a substantive modeling choice rather than a harmless scaling convention.

3.5 Choice probabilities and market shares

Let A_{jt} be the set of consumer attributes for which product j gives the highest utility in market t :

$$A_{jt}(x_t, p_t, \delta_t; \theta_2) = \{(D_i, \nu_i, \varepsilon_i) : u_{ijt} \geq u_{ilt} \forall l = 0, 1, \dots, J_t\}. \quad (6)$$

The model-predicted market share is the mass of consumers in this set:

$$s_{jt}(x_t, p_t, \delta_t; \theta_2) = \int_{A_{jt}} dP^*(D, \nu, \varepsilon). \quad (7)$$

Interpretation.

- The market share is an integral over heterogeneous consumers.
- With Type I extreme-value ε_{ijt} , the idiosyncratic shocks can be integrated analytically.
- With random coefficients, the remaining integration over D_i and ν_i usually requires simulation.

3.6 The simple logit benchmark

If $\theta_2 = 0$, then all consumers have the same α and β . With i.i.d. Type I extreme-value shocks, the model becomes the aggregate logit:

$$s_{jt} = \frac{\exp(x_{jt}\beta - \alpha p_{jt} + \xi_{jt})}{1 + \sum_{k=1}^{J_t} \exp(x_{kt}\beta - \alpha p_{kt} + \xi_{kt})}. \quad (8)$$

The logit price elasticities are

$$\eta_{jkt} = \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}} = \begin{cases} -\alpha p_{jt}(1 - s_{jt}), & j = k, \\ \alpha p_{kt} s_{kt}, & j \neq k. \end{cases} \quad (9)$$

Why the logit is too restrictive.

- Own-price elasticities are largely proportional to own price when shares are small.
- Cross-price elasticities depend only on the market share of the product whose price changes.
- Consumers leaving one product substitute toward all other products in proportion to their market shares.
- The model cannot naturally say that two children's cereals are closer substitutes than a children's cereal and a nutrition cereal unless that is built into the model through richer structure.

3.7 Nested logit and GEV models

Nested logit and other generalized extreme-value models relax the i.i.d. logit substitution pattern by allowing correlation in ε_{ijt} across related products.

- Products are grouped into nests, such as children's cereals, adult cereals, or nutrition cereals.
- Substitution is stronger within a nest than across nests.
- The main advantage is tractability: market shares still have analytic expressions.
- The main limitation is that substitution still depends on a priori group definitions, and within a group substitution remains restrictive.

Interpretation. Nested logit is a useful intermediate case. It improves on simple logit but still forces the researcher to choose nests and cannot flexibly represent continuous distances in product-characteristic space.

3.8 Full random-coefficients logit elasticities

With heterogeneous price sensitivity, individual i 's conditional logit probability is

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J_t} \exp(\delta_{kt} + \mu_{ikt})}. \quad (10)$$

The aggregate share is the average of s_{ijt} over consumers. The price elasticities become

$$\eta_{jkt} = \frac{p_{kt}}{s_{jt}} \begin{cases} - \int \alpha_i s_{ijt} (1 - s_{ijt}) dP^*(D, \nu), & j = k, \\ \int \alpha_i s_{ijt} s_{ikt} dP^*(D, \nu), & j \neq k. \end{cases} \quad (11)$$

Interpretation.

- Own-price elasticity is no longer mechanically tied to price alone.
- If the consumers who buy a product are highly price sensitive, that product can have high own-price elasticity even if its price is low.
- Cross-price substitution is stronger between products that attract similar consumers.
- Similarity is determined by observed characteristics and by how consumer tastes load on those characteristics.

4 Estimation algorithm (all equations + interpretation)

4.1 The tempting but incomplete objective

A direct approach would choose parameters to make predicted shares close to observed shares:

$$\min_{\theta} \|s(x, p, \delta(x, p, \xi; \theta_1); \theta_2) - S\|. \quad (12)$$

The paper explains why this is not the preferred approach.

- All parameters enter nonlinearly, which is computationally expensive.
- More importantly, this objective does not directly handle the endogeneity of prices.
- The structural error should be ξ_{jt} , not just the share prediction error.

4.2 GMM moments

Let Z be a matrix of instruments satisfying

$$\mathbb{E}[Z_{jt} \omega_{jt}(\theta^*)] = 0, \quad (13)$$

where $\omega_{jt}(\theta)$ is the structural error implied by the model. The GMM estimator is

$$\hat{\theta} = \arg \min_{\theta} \omega(\theta)' Z \hat{\Omega}^{-1} Z' \omega(\theta). \quad (14)$$

Interpretation.

- At the true parameters, the structural demand error should be orthogonal to valid instruments.
- If there are more instruments than parameters, GMM chooses the parameters that make the moments as close to zero as possible.
- The weight matrix gives less weight to noisier moments.

4.3 Key inversion step

For a candidate $\theta_2 = (\Pi, \Sigma)$, solve for mean utilities δ_t such that predicted shares match observed shares in each market:

$$s(\delta_t; \theta_2) = S_t, \quad t = 1, \dots, T. \quad (15)$$

Plain English. Given a guess about heterogeneity, find the vector of product mean utilities that rationalizes the observed market shares.

Why this is the key move.

- ξ_{jt} enters utility only through δ_{jt} .
- Once δ_{jt} is recovered, the residual ξ_{jt} can be written like a regression error.
- This transforms a complicated demand system into a nonlinear IV/GMM problem with a structural residual.

4.4 Simulated market shares

For the full random-coefficients model, the share integral is approximated by simulation. Draw n_s consumer types (D_i, ν_i) . Then

$$s_{jt}(p_t, x_t, \delta_t; \theta_2) \approx \frac{1}{n_s} \sum_{i=1}^{n_s} \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{m=1}^{J_t} \exp(\delta_{mt} + \mu_{imt})}. \quad (16)$$

Implementation note.

- The Type I extreme-value shocks are integrated out analytically.
- The remaining heterogeneity is integrated out by simulation or numerical integration.
- More simulation draws improve accuracy but increase computation time.

4.5 Contraction mapping

BLP solve equation (15) using the contraction mapping

$$\delta_{.t}^{h+1} = \delta_{.t}^h + \log S_t - \log s(p_{.t}, x_{.t}, \delta_{.t}^h; \theta_2), \quad h = 0, 1, 2, \dots \quad (17)$$

Stop when $\|\delta_{.t}^{h+1} - \delta_{.t}^h\|$ is below a tolerance level.

Interpretation.

- If predicted share is too low relative to observed share, raise δ_{jt} .
- If predicted share is too high relative to observed share, lower δ_{jt} .
- The algorithm iterates until the model exactly fits market shares for the given θ_2 .

4.6 Recovering the structural error

After the inversion, the residual is

$$\omega_{jt}(\theta) = \xi_{jt}(\theta) = \delta_{jt}(S_t; \theta_2) - x_{jt}\beta + \alpha p_{jt}. \quad (18)$$

Equivalently, since $\delta_{jt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$, equation (18) just solves for the implied unobserved product quality.

How the nonlinear search works.

- Search over nonlinear parameters $\theta_2 = (\Pi, \Sigma)$.
- For each candidate θ_2 , invert shares to recover δ .
- Given δ , estimate the linear parameters $\theta_1 = (\alpha, \beta)$ by IV/GMM.
- Form $\xi(\theta)$, compute moments, and evaluate the GMM objective.
- Choose θ_2 that minimizes the objective; recover θ_1 at that optimum.

4.7 Special cases

- **Logit.** The inversion is analytic:

$$\delta_{jt} = \log S_{jt} - \log S_{0t}.$$

Estimation reduces to IV regression of $\log S_{jt} - \log S_{0t}$ on price and characteristics.

- **Nested logit and GEV.** The inversion is also analytic or simpler because market shares have closed-form expressions.
- **Full random-coefficients logit.** The share integral and inversion are numerical, but the same structural residual and GMM logic apply.

5 Identification and instruments (what makes the estimates credible)

5.1 Informal identification logic

The ideal experiment would randomly assign consumers to different price vectors and observe their choices. With aggregate market data, the model instead uses variation in product prices, characteristics, market shares, choice sets, and demographics across markets.

Key identifying variation.

- Variation in prices helps identify price sensitivity.
- Variation in product characteristics helps identify tastes for characteristics.
- Variation in the choice set helps identify substitution patterns.
- Variation in demographics across markets helps identify how tastes vary with demographics.

5.2 How random coefficients are identified

Consider three products A , B , and C . Suppose A and B are similar in characteristics, while B and C have the same market share. If the price of A rises:

- the simple logit predicts that B and C gain the same amount of share, because they have the same initial share;
- the random-coefficients model allows B to gain more share because it is closer to A in characteristics space.

Interpretation. Observing whether consumers move toward similar products rather than merely large-share products helps identify the parameters that govern heterogeneity.

5.3 Why prices are endogenous

The product characteristic ξ_{jt} is observed by firms and consumers but not by the econometrician. A differentiated-products pricing model generally implies that prices depend on marginal cost and markups. Markups depend on demand, including ξ_{jt} .

$$\text{Cov}(p_{jt}, \xi_{jt}) \neq 0 \tag{19}$$

Consequence. Treating price as exogenous will generally bias the estimated price coefficient and therefore distort markups, merger simulations, welfare calculations, and product-innovation valuations.

5.4 Cost shifters

The textbook instruments for demand are variables that shift marginal cost but are uncorrelated with demand shocks.

- **Strength.** They have a clean economic interpretation.
- **Weakness.** In differentiated-product data, cost data are often unavailable or too coarse to vary at the brand-product-market level.
- **Practical implication.** Cost shifters are attractive when available, but in many applications researchers must rely on other instruments.

5.5 BLP-style characteristics instruments

A widely used approach assumes product locations in characteristics space are predetermined with respect to current demand shocks. Instruments include:

- the product's own observed characteristics, excluding price and other endogenous variables;
- sums of characteristics of other products produced by the same firm;
- sums of characteristics of products produced by rival firms.

Economic logic.

- Rivals' product characteristics affect equilibrium markups and prices.
- If product characteristics are predetermined relative to current demand shocks, they can shift price without being correlated with ξ_{jt} .
- These instruments vary by product, which is especially useful in differentiated-product settings.

Potential problems.

- Observed characteristics may be chosen in response to expected unobserved quality or demand.
- Some product categories are better measured than others, creating correlation between observed and unobserved quality.
- If product characteristics can be changed quickly, they may react to the same shocks as price.
- With brand fixed effects and a constant choice set, these instruments may have little within-brand market variation.

5.6 Hausman cross-market instruments

Another approach exploits panel structure. The price of product j in other markets can instrument for the price of product j in market t .

Logic.

- Prices of the same product across cities are correlated because of common marginal costs.
- If market-specific demand shocks are independent across cities, another city's price is uncorrelated with the local demand residual.

Potential problems.

- National advertising, national promotions, or national health news can create correlated demand shocks across cities.

- Local promotions coordinated across nearby cities can violate cross-market independence.
- These instruments are more plausible when market-specific demand shocks are local and not correlated across markets.

5.7 Brand fixed effects

Nevo emphasizes that product or brand fixed effects should be used whenever possible.

- They absorb time-invariant observed and unobserved product quality.
- They improve the fit of mean utility.
- They reduce the burden on instruments by removing the average component of unobserved quality.

If fixed effects are included, the error becomes the market-specific deviation:

$$\tilde{\xi}_{jt} = \xi_{jt} - \xi_j. \quad (20)$$

Interpretation. Instruments no longer need to be orthogonal to permanent product quality, but they still need to be orthogonal to market-specific demand shocks.

5.8 Recovering taste coefficients with brand fixed effects

With brand fixed effects, coefficients on time-invariant characteristics cannot be estimated in the first-stage demand equation. Nevo explains that they can be recovered by minimum distance.

Let d be the vector of estimated brand fixed effects, X be the matrix of time-invariant product characteristics, and ξ be brand-specific unobserved quality:

$$d = X\beta + \xi. \quad (21)$$

If $\mathbb{E}[\xi \mid X] = 0$, then

$$\hat{\beta} = \left(X'V_d^{-1}X \right)^{-1} X'V_d^{-1}\hat{d}, \quad \hat{\xi} = \hat{d} - X\hat{\beta}, \quad (22)$$

where V_d is the variance-covariance matrix of the estimated brand effects.

Interpretation.

- The first step estimates unrestricted product mean utilities.
- The second step projects those mean utilities onto product characteristics.
- The maintained assumption is that the remaining brand quality is orthogonal to observed characteristics.

5.9 Bottom line on credibility

- The estimator is not credible simply because it is sophisticated.
- Credibility comes from the economic argument for instruments, the richness of market variation, and the robustness of results to alternative instrument sets and market-size definitions.
- The most important practical advice in the paper is not to let computational complexity obscure the basic IV problem.

6 Application, tables, and figures

6.1 Application setting

The paper includes a teaching application motivated by scanner data for ready-to-eat cereal. The data are simulated rather than real and are intended to help readers learn the method.

- There are **24 brands** of a differentiated product.
- There are **47 cities** and **2 quarters**, giving **94 markets**.
- With 24 brands per market, the sample has **2,256 product-market observations**.
- The two product characteristics are **Sugar** and **Mushy**.
- **Mushy** equals one if the product becomes soggy in milk.
- Demographics include log income, log income squared, age, and an indicator for being a child.
- The simulation uses $n_s = 20$ consumer draws per market.

6.2 Table I: results from the full model

Read: Table I reports estimates of mean tastes, unobserved taste dispersion, and interactions between tastes and demographics.

Key numbers.

- The price mean is -32.433 with standard error 7.743 .
- The GMM objective is **14.9** with **7** degrees of freedom.
- The sample has **2,256** product-market observations.
- The estimates include brand and time dummy variables.

Interpretation.

- The mean price coefficient is negative, as demand theory requires.
- Demographic interactions are economically meaningful even when statistical precision is limited.
- The estimated standard deviations are small, suggesting that in this demonstration data most heterogeneity is captured by observed demographics rather than by the assumed normal random coefficients.

TABLE I.
RESULTS FROM THE FULL MODEL

Variable	Means	Standard Deviations	Interactions with Demographic Variables			
	β	σ	Income	IncomeSq	Age	Child
Price	-32.433 (7.745)	1.848 (1.075)	16.598 (172.334)	-0.659 (8.955)	—	11.625 (5.207)
Constant	-1.841 ^a (0.258)	0.377 (0.129)	3.089 (1.213)	—	1.186 (1.016)	—
Sugar	0.148 ^a (0.258)	0.004 (0.012)	-0.193 (0.005)	—	0.029 (0.036)	—
Mushy	0.788 ^a (0.013)	0.081 (0.205)	1.468 (0.697)	—	-1.514 (1.103)	—
GMM objective (Degrees of freedom)			14.9 (7)			

Based on 226 observations. Except where noted, parameters are GMM estimates. All regressions include brand and time dummy variables. Asymptotically robust standard errors are given in parentheses.
^aEstimates from a minimum-distance procedure.

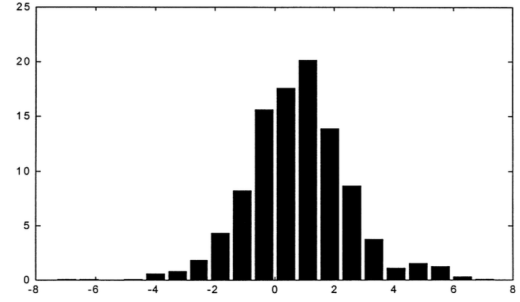


FIGURE 1. FREQUENCY DISTRIBUTION OF TASTE FOR SOGGINESS

6.3 Figure 1: distribution of taste for sogginess

Read: Figure 1 plots the implied cross-consumer distribution of the coefficient on the Mushy characteristic.

- Most consumers have positive valuation for sogginess.
- About **31%** of consumers instead prefer a crunchy cereal.
- The figure illustrates why mean coefficients are not enough: the average consumer likes Mushy, but a substantial minority dislikes it.

Why it matters. Random coefficients turn a single taste coefficient into a full distribution. That distribution determines which products are substitutes for which consumers.

6.4 Figure 2: distribution of the price coefficient

Read: Figure 2 plots the implied distribution of individual price sensitivity.

- The distribution is not normal because it is partly driven by the empirical distribution of demographics.
- The model in principle could generate positive price coefficients in the tail, but the estimates in this application do not.
- Children and higher-income consumers are estimated to be less price sensitive.

Why it matters. The distribution of price sensitivity is central for markups, elasticities, merger simulations, and welfare calculations.

6.5 Table II: own- and cross-price elasticities

Read: Table II reports median own- and cross-price elasticities across the 94 markets. Entry (i, j) is the percent change in market share of brand i from a one-percent price increase for brand j .

Key own-price elasticities shown in the table.

- Brand 1: -2.1689.
- Brand 2: -3.4505.

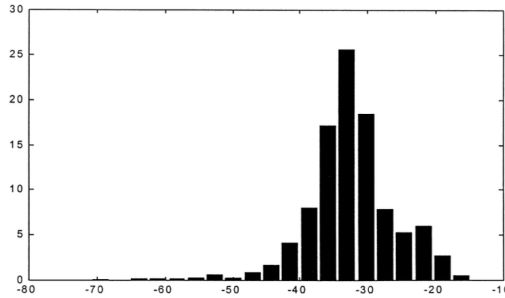


FIGURE 2. FREQUENCY DISTRIBUTION OF PRICE COEFFICIENT

- Brand 3: -2.9208 .
- Brand 4: -3.2709 .
- Brand 5: -5.5116 .
- Brand 14: -3.8393 .
- Brand 24: -4.1215 .

Interpretation.

- Own-price elasticities are all negative, as expected.
- Cross-price elasticities vary substantially across products.
- Brands with similar characteristics have stronger substitution patterns.
- Brands 14 and 15 have identical observed characteristics, so their cross-price elasticities are essentially identical.
- The ratio of maximum to minimum cross-price elasticities within a column ranges from about **3** to **9**. Under simple logit this ratio would be **1**.

Why it matters. This is the clearest diagnostic that the random-coefficients model has escaped the restrictive logit substitution pattern.

7 Implementation checklist

7.1 Step-by-step recipe

1. Define markets and products.
2. Measure product quantities and prices.
3. Choose total market size and compute inside and outside shares.
4. Collect product characteristics that plausibly drive substitution.
5. Decide which characteristics have random coefficients.
6. Bring in demographic distributions if available.

TABLE II.
MEDIAN OWN- AND CROSS-PRICE ELASTICITIES

Brand	Characteristics		Elasticity									
	Sugar	Mushy	1	2	3	4	5	14	15	19	24	
1	2	1	-2.1689	0.0562	0.2429	0.1161	0.0645	0.0596	0.0611	0.0759	0.2375	
2	18	1	0.1119	-3.4555	0.1302	0.0924	0.1905	0.2305	0.2606	0.4380	0.1145	
3	4	1	0.2715	0.0674	-2.9208	0.1083	0.0709	0.0627	0.0684	0.0944	0.2237	
4	3	0	0.1868	0.0736	0.1501	-3.2709	0.1470	0.1557	0.1564	0.1037	0.2670	
5	12	0	0.0865	0.1540	0.0847	0.1336	-5.5116	0.2652	0.3085	0.2988	0.1715	
6	14	0	0.0699	0.1642	0.0698	0.1123	0.2518	0.3628	0.3971	0.3513	0.1369	
7	3	1	0.2828	0.0606	0.2290	0.1035	0.0648	0.0563	0.0598	0.0865	0.2225	
8	4	0	0.1743	0.0799	0.1476	0.1622	0.1534	0.1620	0.1656	0.1180	0.2585	
9	14	0	0.0735	0.1704	0.0717	0.1128	0.2596	0.3586	0.3966	0.3443	0.1333	
10	1	0	0.2109	0.0643	0.1682	0.1685	0.1224	0.1199	0.1298	0.0817	0.2809	
11	11	0	0.0967	0.1556	0.0890	0.1377	0.2426	0.3322	0.3427	0.2899	0.1769	
12	4	1	0.2711	0.0691	0.2339	0.1121	0.0735	0.0662	0.0704	0.0985	0.2260	
13	3	1	0.2627	0.0597	0.2290	0.0983	0.0632	0.0554	0.0582	0.0866	0.2189	
14	13	0	0.0834	0.1825	0.0786	0.1247	0.2697	-3.8393	0.3022	0.3380	0.1600	
15	13	0	0.0889	0.1763	0.0814	0.1236	0.2491	0.3542	-4.9862	0.3383	0.1582	
16	16	1	0.1480	0.2282	0.1323	0.0873	0.1447	0.1442	0.1702	0.1264	0.1864	
17	10	0	0.1138	0.1399	0.1005	0.1466	0.2172	0.2867	0.2912	0.2498	0.1875	
18	3	0	0.1900	0.0717	0.1542	0.1656	0.1362	0.1508	0.1578	0.1026	0.2660	
19	20	1	0.0944	0.2691	0.1130	0.0771	0.2163	0.2618	0.3107	-3.3335	0.1013	
20	7	0	0.1420	0.0999	0.1270	0.1546	0.1791	0.2074	0.2080	0.1628	0.2251	
21	14	0	0.0773	0.1706	0.0741	0.1160	0.2534	0.3573	0.3794	0.3530	0.1416	
22	6	0	0.1500	0.0950	0.1283	0.1602	0.1797	0.2096	0.2085	0.1530	0.2211	
23	12	0	0.0854	0.1561	0.0815	0.1307	0.2599	0.3484	0.3753	0.3154	0.1608	
24	0	0	0.2249	0.0548	0.1731	0.1619	0.1110	0.1044	0.1114	0.0733	-4.1215	
Outside good			0.0803	0.2433	0.0396	0.0552	0.2279	0.3349	0.3716	0.3896	0.0399	

Cell entries ϵ_{ij} where i indexes row and j columns, give the percent change in market share of brand i with a one-percent change in price of j . Each entry represents the median of the elasticities from the 94 markets.

7. Choose instruments for endogenous prices.
8. Include brand fixed effects when enough markets are available.
9. Draw simulated consumer types and hold draws fixed during optimization.
10. For each candidate θ_2 , compute predicted shares by simulation.
11. Use the contraction mapping to invert observed shares into δ .
12. Recover ξ , compute GMM moments, and minimize the objective.
13. Use estimates to compute elasticities, markups, welfare, merger effects, or new-product values.

7.2 Common pitfalls

- **Bad market-size definition.** If the outside share is implausible, elasticities can be badly distorted.
- **Weak product characteristics.** If characteristics do not describe the relevant dimensions of substitution, the model may not improve much over logit.
- **Invalid instruments.** Sophisticated nonlinear estimation cannot fix invalid instruments.
- **Too few markets.** Multiple markets help identify heterogeneity, especially demographic interactions and random-coefficient variance.
- **Overinterpreting means.** The economically relevant object is often the full distribution of tastes, not just the average coefficient.
- **Ignoring fixed effects.** Without brand fixed effects, permanent unobserved product quality can contaminate price estimates.

8 Extensions discussed in the paper

8.1 Supply side

The paper uses supply-side reasoning mainly to motivate instruments. But a researcher can jointly estimate demand and supply by adding supply-side moment conditions to the GMM objective.

- **Benefit.** Joint estimation can improve efficiency and recover marginal costs or markups.
- **Cost.** It requires stronger assumptions about firm conduct, ownership, cost functions, and equilibrium pricing.
- **Use case.** Merger simulations and market-power measurement often require a supply side.

8.2 Consumer-level data

If individual purchase data are available, one can use maximum likelihood, simulated method of moments, or hybrid macro-micro moment approaches.

- Consumer-level data can help estimate heterogeneity more directly.

- Aggregate moments can still be used to discipline market shares.
- Price endogeneity may remain even with household-level data if product-market unobservables are omitted.

8.3 Alternative demand systems

Multi-level demand systems are an alternative to random-coefficients logit. They use separability and aggregation assumptions to model demand at different levels, such as product category, segment, and brand.

- They can be flexible when the number of brands per segment is small.
- They become difficult when there are many brands or unclear segments.
- They require a priori segmentation, while random-coefficients logit uses product characteristics to generate substitution patterns.

8.4 Dynamics

The paper also notes links to dynamic demand and dynamic supply models.

- Dynamic firm models can endogenize product characteristics and innovation.
- Dynamic demand models can incorporate learning, switching costs, advertising effects, or purchase history.
- Such models are often better suited to consumer-level or high-frequency data.

9 Short summary

- **Problem.** Differentiated-product demand estimation needs flexible substitution patterns, but unrestricted demand systems are too high-dimensional.
- **Simple logit.** Solves dimensionality but imposes unrealistic substitution: products gain consumers in proportion to market share.
- **Random-coefficients logit.** Allows consumers to differ in price sensitivity and tastes for characteristics, generating substitution toward similar products.
- **Main econometric insight.** Invert market shares to recover mean utilities, define the structural residual as unobserved product quality, and estimate using IV/GMM.
- **Main identification issue.** Price is endogenous because firms observe product quality shocks when setting prices.
- **Main instruments.** Cost shifters, BLP-style characteristics instruments, and Hausman cross-market price instruments.
- **Practical lesson.** Use brand fixed effects when possible, choose market size carefully, collect product characteristics that capture substitution, and test robustness to instruments.

- **Core output.** Elasticities, markups, diversion patterns, merger simulations, and welfare calculations that are more credible than those from simple logit.

10 Conclusion

Core conclusion. Nevo's guide explains how random-coefficients logit models make differentiated-product demand estimation practical with aggregate data. The method converts observed prices and market shares into a structural demand system with heterogeneous consumers and endogenous prices.

Economic intuition. Consumers choose bundles of characteristics. If some consumers value the same characteristics, they will view similar products as close substitutes. Random coefficients let the data reveal these substitution patterns, while the BLP inversion lets the researcher recover unobserved product quality and estimate the model with instruments.

Main takeaway. The model is powerful because it is both economically rich and empirically feasible. But the credibility of the estimates still rests on the basics: good data, sensible market definitions, relevant product characteristics, and defensible instruments.